PERFORMANCEINTELLIGENCE

ForecastIQ

Probabilistic revenue planning and risk-aware budget guidance — 30, 60, and 90-day outlook

AUDITDATE	DATAQUALITY	RISKCLASSIFICATION	PLATFORMMODE
2026-07-15	**100.0/100**	**High Risk**	**Offline Utility**

1 Executive Summary

KEYTAKEAWAY

Over the full historical timeline, ForecastIQ has audited $2.18M in digital marketing spend yielding $11.10M in revenue (Portfolio ROAS: 5.09x). Our 90-day probabilistic ensemble projects sustained top-line stability with a P50 expected capture of $0.96M. Meta Ads continues to function as your high-efficiency anchor at 8.44x ROAS, while Google Ads shows the strongest recent spend momentum for scalable incremental demand.

2 Multi-Horizon Revenue & ROAS Projections

The following forecasts are derived from a weighted ensemble (Prophet, XGBoost, LightGBM, CatBoost). P10 is a conservative floor, P50 the expected baseline, and P90 an upside scenario — not point estimates, but a probability range meant to be planned against.

Planning Window	Metric	P10 (Conservative)	P50 (Expected)	P90 (Upside)
30 Days	**Revenue**	$96,021	**$318,369**	$540,717
30 Days	**ROAS**	1.64x	**5.43x**	9.22x
60 Days	**Revenue**	$266,335	**$640,279**	$1,014,222
60 Days	**ROAS**	2.25x	**5.40x**	8.55x
90 Days	**Revenue**	$456,965	**$963,809**	$1,470,653
90 Days	**ROAS**	2.56x	**5.41x**	8.25x

3 Forecast Accuracy: Offline Backtest vs. Naive Baseline

3-fold rolling-origin

To check whether the uncertainty bands above are meaningful rather than arbitrary, the ensemble was evaluated with a rolling-origin backtest: retrained on progressively larger historical windows and scored against forecast periods it had not seen, then compared to a naive baseline (a flat projection of the trailing 30-day daily average). Figures below are computed at the Overall (portfolio) level, where percentage-based accuracy is stable; Campaign-level rows are noisier and are addressed separately below.

Window	Metric	Model MAPE	Naive Baseline MAPE	P10–P90 Coverage	Result
30 Days	**ROAS**	20.0%	23.8%	100%	**Beats baseline** (+16.1%)
30 Days	**Revenue**	18.5%	17.6%	100%	**Trails baseline** (-5.1%)
60 Days	**ROAS**	9.4%	13.3%	100%	**Beats baseline** (+29.4%)
60 Days	**Revenue**	17.4%	18.4%	100%	**Beats baseline** (+5.5%)
90 Days	**ROAS**	16.8%	18.0%	100%	**Beats baseline** (+6.7%)
90 Days	**Revenue**	6.1%	6.6%	100%	**Beats baseline** (+6.9%)

ROAS forecasts beat the naive baseline at all three horizons (30-day, 60-day, 90-day). Revenue forecasts beat the naive baseline at 60-day (+5.5%), 90-day (+6.9%) and trail it at 30-day (-5.1%). We're disclosing this rather than omitting it. Tracing the gap by decomposing the forecast into its components found that the model's own recency-anchoring step — which blends the raw ensemble output with a trailing revenue baseline to stabilize it — was weighting a stale 90-day average too heavily relative to the more recent 30-day level the naive baseline itself uses, overshooting by 18–23% APE on its own before the underlying model even contributed. Re-weighting that anchor toward the same recency the naive baseline uses cut the Revenue MAPE gap by roughly 60–75% across all three horizons. Two further factors widen the remaining gap: (1) the earliest backtest origin trains on only ≈ 4 months of history — well short of what Prophet's seasonal component needs to calibrate reliably, and (2) the dataset contains a genuine 5–10x November–December demand spike that appears in only two historical instances across the ≈ 2.5-year dataset, so any forecast horizon crossing that window is working with minimal precedent — a constraint that affects any forecasting method, not just this one.

KNOWNLIMITATIONS

Forecast accuracy improves with training history and is weakest for the earliest, data-sparse backtest origin. November–December forecasts carry wider uncertainty given only two observed instances of the holiday demand spike in the historical record. Campaign-level forecasts (excluded from the table above) inherit materially higher variance than portfolio-level ones, since individual campaign lifecycles and end-dates aren't modeled as an explicit signal.

4 Channel Performance & Contribution Breakdown

Treating existing channel attribution as the source of truth, here is the forward 60-day expected contribution and efficiency analysis across your primary paid acquisition channels.

Acquisition Channel	Metric	P10	P50 (Expected)	P90
Bing Ads	Revenue	$609	**$12,174**	$49,397
Bing Ads	ROAS	0.09x	**1.80x**	7.30x
Google Ads	Revenue	$28,631	**$572,625**	$1,305,332
Google Ads	ROAS	0.28x	**5.66x**	12.90x
Meta Ads	Revenue	$4,225	**$84,509**	$411,856
Meta Ads	ROAS	0.28x	**5.61x**	27.33x

5 Budget Optimizer Recommendations

An Optuna-driven search allocated a monthly budget of **$100,000** to maximize total revenue subject to a target ROAS floor of **4.5x**. The search assumes diminishing marginal returns per channel — pushing more spend into one channel yields progressively less revenue per dollar — so the optimizer balances the split across all channels rather than concentrating on whichever looks best in isolation.

Channel	Recommended Spend	Expected Revenue	Expected ROAS	Spend Share
Bing Ads	$1,097	$5,139	**4.69x**	1.1%
Google Ads	$21,591	$121,252	**5.62x**	21.7%
Meta Ads	$76,955	$444,241	**5.77x**	77.2%
OPTIMIZED TOTAL	**$99,642**	**$570,632**	**5.73x**	**100.0%**

READINGTHISTABLE

Meta Ads receives the largest share of this recommendation (77% of budget) at an expected 5.77x ROAS — as spend rises, revenue increases but marginal ROAS typically falls. The optimizer's job is to find the allocation that protects your ROAS floor while still increasing revenue — a business decision, not just a bigger number.

6 Risk Intelligence & Growth Action Plan

The overall portfolio risk classification is **High Risk** (risk score: 70.3/100). Below are the specific operational risks identified and recommended mitigation actions.

Risk Factor	Score	Status	Mitigation
Revenue Volatility	100.0/100	**Monitor**	Utilize automated pacing tools to smooth out mid-week revenue depressions and maintain consistent top-of-funnel impression share.
Channel Dependency	58.5/100	**Concentrated**	Google Ads represents 83.5% of revenue capture. Begin scaling top-of-funnel prospecting on your other channels to establish a strong secondary growth pillar.
ROAS Instability	85.4/100	**Volatile**	Enforce rigorous Target CPA / Target ROAS bid floor rules across all Bing Ads generic and non-brand campaigns to prevent runaway spend.
Data Quality Audit	0.0/100	**Excellent**	Validation engine verified a 100.0/100 schema score. Ensure UTM tracking tags are consistently and correctly appended across all ingested channel campaigns.

ForecastIQ — built for the NetElixir AIgnition 3.0 Hackathon Challenge.

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